

Ahib Afnan Siam

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Professional Summary

AI Software Engineer specializing in production-grade **LLM systems, RAG, NL2SQL, semantic search, and AI-powered operations**. Built enterprise AI platforms operating over **50M+ records** and large-scale CV datasets, including natural-language analytics and semantic candidate matching systems. Strong in Python, FastAPI, Oracle VECTOR, retrieval and reranking pipelines, and end-to-end AI system design and deployment.

Experience

AI Software Engineer (Sub Assistant Manager) — MIS

PRAN-RFL Group

Dhaka, Bangladesh

November 2025 – Present

- Built **Uttoron**, a natural-language-to-SQL AI assistant using FastAPI, ChromaDB, and Oracle with LLM-driven query generation and analytics
- Designed schema-aware RAG pipelines enabling real-time SQL execution over **50M+ records**
- Automated MIS reporting workflows, improving data accessibility and decision-making speed
- Developed a full-stack **AI Recruitment Automation system** processing **10,000+ CVs** with semantic job matching
- Built a semantic candidate matching engine using Oracle VECTOR search + MiniLM reranking, improving ranking accuracy
- Designed hybrid vector + reranker scoring with an explainable 0–100 candidate ranking system
- Built an async CV processing pipeline with real-time tracking and job-wise candidate ranking
- Reduced manual screening effort by **60%**
- Engineered an **AI meeting agent for Google Meet & Microsoft Teams** using Playwright and Whisper for auto-join, participant tracking, transcription, AI summaries, and PDF report generation

AI Software Engineer (Intern) — MIS

PRAN-RFL Group

Dhaka, Bangladesh

June 2025 – November 2025

- Contributed to development of an NL→SQL AI assistant by implementing schema-aware query generation and validation, improving query accuracy
- Optimized RAG pipeline performance and assisted in on-premise deployment, enhancing system reliability and reducing report generation time

Full Stack Engineer Intern

Universe IT Institute (UITI)

Dhaka, Bangladesh

December 2024 – May 2025

- Developed AI-powered and full-stack web applications using Python, FastAPI, React, Next.js, Node.js, and PHP
- Built and integrated REST APIs, ML-powered features, and frontend-backend functionalities for scalable web applications

Education

BRAC University, Dhaka

Bachelor of Science in Computer Science and Engineering

July 2020 - January 2025

Projects

PRAN-RFL AI Assistant - Uttoron

Enterprise Production System — Internal Network Deployment

- Built a natural-language-to-SQL AI assistant for enterprise analytics and reporting
- Enabled real-time query generation and insights using LLMs with schema-aware RAG pipelines
- **Tech Stack:** Python, FastAPI, React, ChromaDB, Sentence Transformers, Oracle, LLM APIs (Ollama, OpenRouter)

Hire360 - PRAN-RFL Group

• Full-Stack AI Recruitment Intelligence Platform

hire360.prangroup.com 

- Built an enterprise AI recruitment platform for CV ingestion, candidate search, matching, shortlisting, and hiring workflows
- Developed async CV processing for PDF/DOCX extraction, profile structuring, bulk uploads, and failure tracking
- Implemented hybrid candidate-job matching using **Oracle VECTOR retrieval + MiniLM reranking** with structured filtering
- Designed separate CV, matching, and LLM workers for scalable background processing and AI-assisted recruiter review

Research

Transitioning to Involutional Neural Network for Resource Efficient Federated Plant Disease Classification

Under Review — Computers and Electronics in Agriculture

- Investigated resource-efficient federated plant disease classification using Involutional Neural Networks, with comparative evaluation against CNN-based architectures

Technical Skills

Languages: Python, TypeScript, JavaScript (ES6+), C/C++, SQL, PHP, C# (basic)

Machine Learning: Scikit-learn, Supervised & Unsupervised Learning, Classification, Regression, Clustering, Feature Engineering, Model Evaluation, Cross-Validation, Hyperparameter Tuning, Ensemble Learning

Deep Learning & Computer Vision: PyTorch, CNNs, RNNs, LSTM, GRU, Transfer Learning, Vision Transformers (ViT), Image Classification, Object Detection, OpenCV, Federated Learning

Generative AI & Information Retrieval: RAG, Semantic Search, Vector Search, Cross-Encoder Reranking, LLM Integration (Ollama, OpenRouter, Groq), FAISS, Sentence Transformers, NL2SQL, Agentic Workflows

Frameworks & Libraries: FastAPI, Node.js, Laravel, React, Next.js, Tailwind CSS, Bootstrap, Pydantic, Chart.js

Databases & Infrastructure: Oracle Database / Oracle 26ai (VECTOR), MySQL, SQLite, ChromaDB, Docker, APScheduler

Developer Tools & Concepts: Git, GitHub, REST APIs, Unit Testing (pytest/Jest), Debugging, Code Reviews, OOP, Data Structures & Algorithms, Authentication & Authorization, Full-Stack AI Development

AI Engineering: Cursor, Claude, ChatGPT, TRAE AI Agent, Qoder, Prompt Engineering, Context Engineering, LLM Debugging

LeetCode: 500+ problems solved — leetcode.com/u/Ahib.Afnan.Siam

HackerRank: 100+ problems solved — hackerrank.com/ahibafnan99

Achievements & Certifications

- **Top 5 (Worldwide)** in Mission OZ 2022 — Space Teams - [View certificate](#) 

2022 - 2023

- **Winner** of Project Showcasing at Mind Sparks 2023, Ahsanullah University of Science And Technology (AUST)

- **Certificate of Accomplishment in Software Engineering**

October 2024

HackerRank — [Credential ID: CF5B58274C39](#) 

- **IELTS Score: 7.5**

5th December 2025